TIC 86498032

Eclipsing binary candidate dossier - Star Vetter

THIN MARGIN

```
BLS EPHEMERIS
  Period       1.229008 d  (ALIASED, see below)
    -> corrected period: 2.458016 d
  Epoch (t0)   2770.1778 BTJD
  Duration     0.0840 d
  Depth        0.784 %  (7,838 ppm)
  SNR          156.0

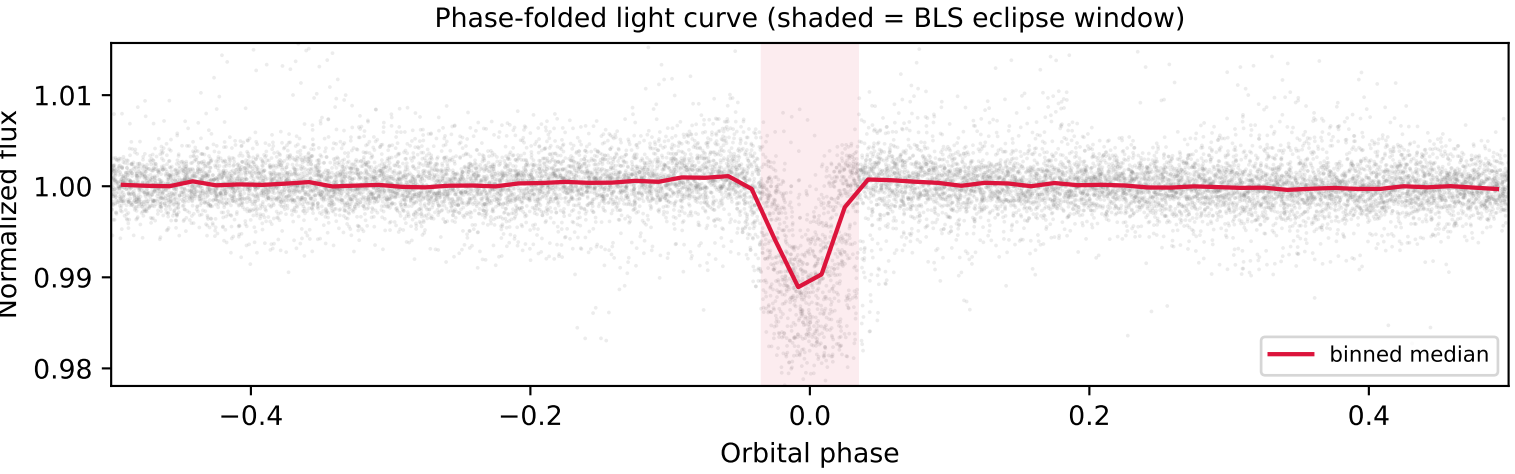
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     42  (gate >= 3)
  Robust depth ratio    1.04  (gate >= 0.50)
  Robust depth sigma    96.8  (gate >= 7.0)
  Odd/even z-score      17.41  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

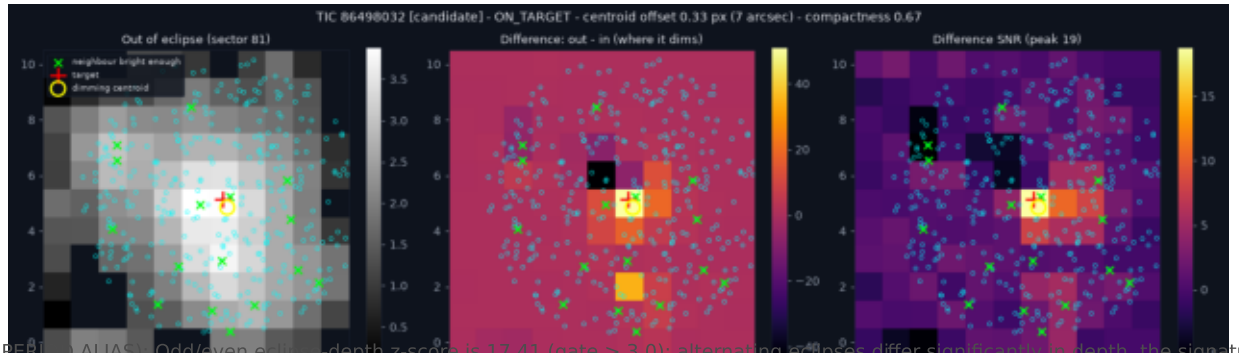
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 81    light-curve dip in this sector: 96.2 sigma    t0 refined by -0.0075 phase
  Difference-image peak SNR   18.77   (gate >= 5.0)      Compactness   0.67   (gate >= 0.50)
  Dimming centroid offset     0.33 px (7 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 86498029
    0.42 px away, dTmag +4.65 - farther from the dimming than the target
```





Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 17.41 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 2.458016 d, in any write-up.
Caveat (THIN MARGIN): The ON_TARGET call rests on a neighbour-competition margin of just 0.087 px (dimming centroid 0.330 px from target vs 0.417 px from the nearest bright-enough neighbour) - inside diag_pixel.py's own stated centroid systematic of 0.10-0.3 px. TESS's resolution cannot confidently separate this star from that neighbour.